Parametric Survival Analysis
Veteran Affairs Lung Cancer Dataset

Group 1

Aline

Anicet

Mireille

Sidoine

Sandra

Group 1 GitHub Repository

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AIMS | African Institute for Mathematical Sciences RWANDA

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Introduction

Introduction

The Veterans' Administration Lung Cancer Study is a randomized clinical trial that investigated survival among patients with advanced lung cancer under two treatment regimens. The dataset is widely used in survival analysis because it contains patient follow-up time, censoring information, treatment group, and important clinical characteristics such as age, cell type, prior therapy, and Karnofsky performance score. In this project, the dataset provides a practical basis for comparing parametric survival models and understanding patient survival patterns over time [1].

Dataset and Objective

The Veteran Affairs lung cancer dataset description

- **time**: survival time
- **status**: event indicator
- **trt**: treatment group
- **celltype**: cancer type
- **karno**: Karnofsky performance score
- **age**: age
- **prior**: prior therapy

Objective

- model survival using parametric models,
- compare models using log-likelihood and AIC,
- select the best-fitting model,
- interpret survival and hazard over time.

Data Summary

Table 1: Basic statistics for key variables

Variable	Total N	Missing N	levels	Control (N=69)	Test therapy (N=68)	Total	p
Age (years)	137 (100.0)	0	Mean (SD)	57.5 (10.8)	59.1 (10.3)	58.3 (10.5)	0.373
Karnofsky score	137 (100.0)	0	Mean (SD)	59.2 (18.7)	57.9 (21.4)	58.6 (20.0)	0.711
Months from diagnosis	137 (100.0)	0	Mean (SD)	8.7 (8.8)	8.9 (12.3)	8.8 (10.6)	0.893
Survival time (days)	137 (100.0)	0	Mean (SD)	115.1 (112.7)	128.2 (193.8)	121.6 (157.8)	0.631
Cell type	137 (100.0)	0	Squamous	15 (21.7)	20 (29.4)	35 (25.5)	0.070
			Small cell	30 (43.5)	18 (26.5)	48 (35.0)	
			Adeno	9 (13.0)	18 (26.5)	27 (19.7)	
			Large	15 (21.7)	12 (17.6)	27 (19.7)	
Prior therapy	137 (100.0)	0	No	48 (69.6)	49 (72.1)	97 (70.8)	0.851
			Yes	21 (30.4)	19 (27.9)	40 (29.2)	
Death (event)	137 (100.0)	0	Censored	5 (7.2)	4 (5.9)	9 (6.6)	1.000
			Death	64 (92.8)	64 (94.1)	128 (93.4)	

Sample Size Data

Table 2: Sample size characteristics

Sample Size	Events	Censored	Mean Time	Min Time	Max Time	Mean Age	Mean Karno
137	128	9	121.63	1	999	58.31	58.57

All p-values are greater than 0.05 as seen in table 1, indicating that there are no statistically significant differences between the control and test therapy groups for any of the variables.

The dataset contains 137 patients, with 128 deaths and 9 censored observations. The average survival time is 121.63 days, while the mean age and mean Karnofsky score are 58.31 years and 58.57, respectively.

Minimum vs Maximum

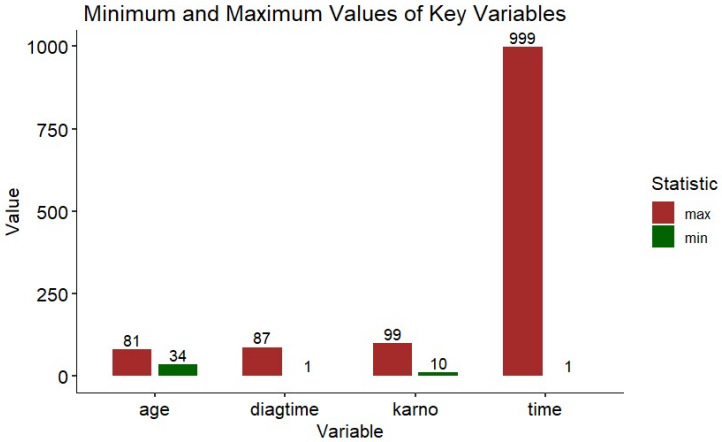

Figure 1: Minimum and maximum for each key variables

Survival times distribution

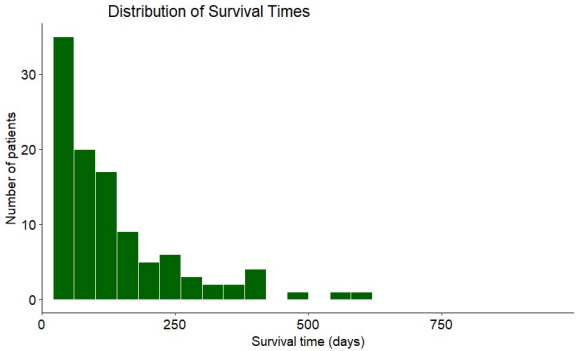

Figure 2: Survival times distribution

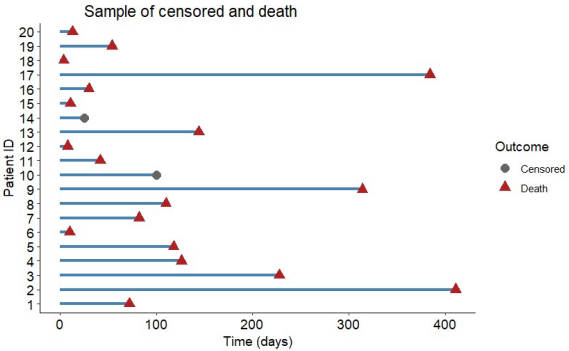

Figure 3: Death and censored through time

The figure 2, Most patients died within the earlier follow-up period, while a small number survived much longer, creating a right-skewed pattern. Figure 3 shows that death did not happen at the same time for all patients, and some patients were still alive at the end of follow-up.

AIC, Loglik and MLE estimates of models fitted

Table 3: AIC, Loglik and MLE estimates of models fitted

Model	LogLik	AIC	MLE Estimates
Exponential	-751.2212	1504.442	$\lambda = 0.0077$
Weibull	-748.0912	1500.182	$k = 0.8521$ $\lambda = 120.6804$
Log-normal	-749.4740	1502.948	$\mu_{log} = 4.1577$ $\sigma_{log} = 1.3783$
Log-logistic	-750.2658	1504.532	$k = 1.2679$ $\lambda = 67.987$
Gompertz	-747.7933	1499.587	$k = -0.0015$ $\lambda = 0.0094$
Gamma	-749.1216	1502.243	$k = 0.8095$ $\lambda = 0.0062$
Generalized Gamma	-746.4708	1498.942	$a = 7.5774$ $d = 1.394$ $p = 0.4511$

Table 3 presents the parametric survival models fitted to the Veteran Affairs lung cancer data, where the outcome of interest was time to death.

Model AIC, Loglik and MLE Interpretation

- **Exponential:** $\lambda = 0.0077$. The hazard is constant over time. This model has the high AIC and gives the poor fit.
- **Weibull:** $k = 0.85 < 1$. The hazard is higher at the beginning and decreases over time. This model improves the fit.
- **Log-normal:** $\sigma_{\log} = 1.38$. The hazard increases first and then decreases. This model shows a moderate fit.
- **Log-logistic:** $k = 1.27$. The hazard rises early, reaches a peak, and then decreases. This model is not among the best.
- **Gompertz:** $k = -0.0015$. The hazard slightly decreases over time. This model shows a good fit.
- **Gamma:** $k = 0.81$. The hazard changes over time. This model has a moderate fit.
- **Generalized Gamma:** $p = 0.45$. This model captures different hazard patterns and has the lowest AIC, giving the best fit.

Model Selection

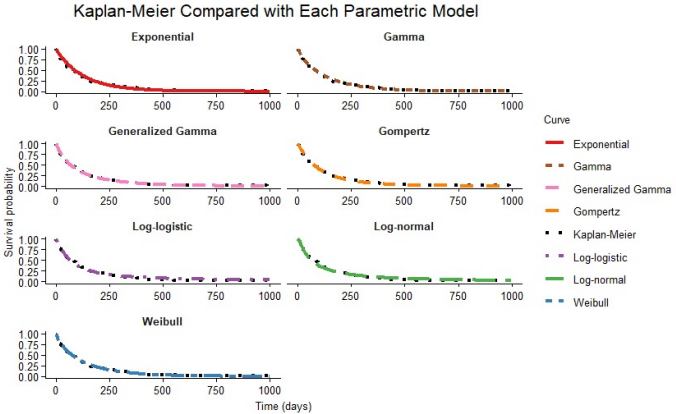

Figure 4: Survival curves for each model to compare fit

The Generalized Gamma model was chosen as the best-fitting model since it had the lowest AIC (1498.942), slightly lower than the Gompertz model (1499.587), and its survival curve closely matched the Kaplan–Meier curve [3], [2].

Life Functions of the Best Model

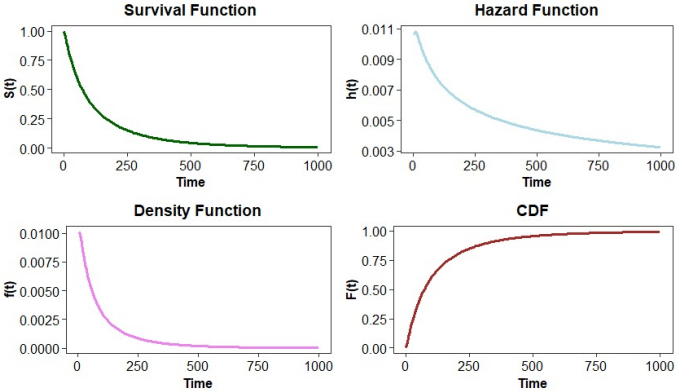

Figure 5: life functions for Generalized Gamma model.

Variance and mean

Table 4: Variance and mean of generalized gamma model

Model	a	d	p	Mean	Variance
Generalized Gamma	7.5774	1.394	0.4511	134.0226	33751.62

- The mean survival time is about 134 days, and the large variance (33751.62) shows high variability in patients' survival times.
- Figure 5 shows that the survival curve declines gradually over time, reflecting the scale parameter $a = 7.5774$, which indicates that survival times are spread over a wide time range.
- The decreasing hazard curve reflects the shape parameter $d = 1.394$, showing that the risk of death is not constant over time.
- The density and CDF indicate many early events and some much later events, reflecting the exponent parameter $p = 0.4511$, which suggests a right-skewed and flexible survival distribution.

By Group model curves

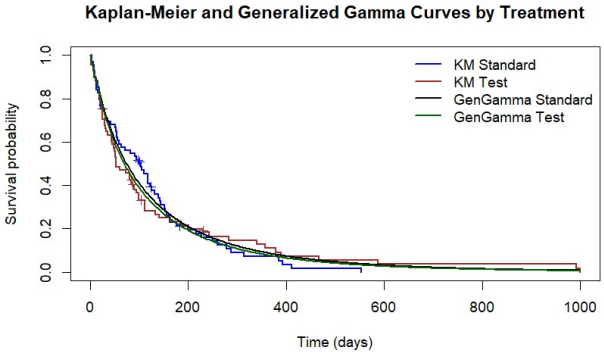

Figure 6: Survival curves by group

Survival declined rapidly in both treatment groups, with no clear visual difference between standard and test therapy. The generalized gamma model closely matched the Kaplan–Meier curves, indicating good model fit.

Model Diagnostic

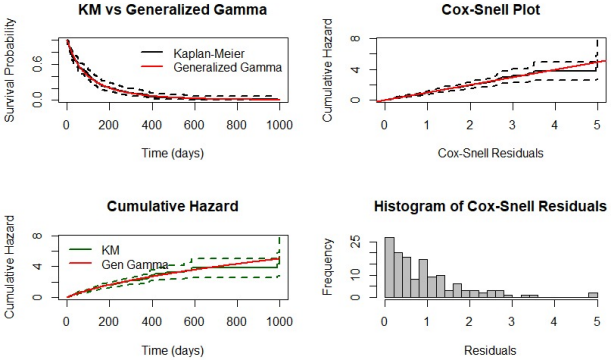

Figure 7: Generalized gamma model diagnostic

- Survival and cumulative hazard curves closely follow the Kaplan–Meier estimates. The Cox–Snell residual plot is approximately linear and the residuals are distributed, indicating an adequate generalized gamma model fit.

Discussion and Public Implications

Discussion

- Most patients experienced the event early, with a few surviving much longer, showing a right-skewed survival pattern.
- The generalized gamma model provided the best fit, capturing the variability and changing risk over time.
- Survival decreases gradually, while the hazard declines over time, indicating higher risk at early stages.

Limitations

- Small sample size (137 patients).
- No multivariable modeling.

Public Implications

- Early stages require close monitoring due to higher risk of death.
- Differences in survival suggest the need for personalized patient care.

Conclusion

Conclusion

- The dataset shows high variability in survival times, with many patients experiencing early events.
- Among the models compared, the generalized gamma model provided the best fit to the data.
- The results indicate that survival decreases over time and the risk of death is not constant.
- Further work is needed to investigate how patient characteristics influence survival and improve model accuracy.

References

[1] John D. Kalbfleisch and Ross L. Prentice. *The Statistical Analysis of Failure Time Data*. Wiley, New York, 1980.

[2] Vinay Prasad. How to read kaplan-meier plots. `https://www.youtube.com/watch?v=W5tDrP2mGLA`, 2024. YouTube video, Vinay Prasad MD MPH, 195K subscribers, 51K views.

[3] Terry M. Therneau. *A Package for Survival Analysis in R*, 2024. R package version 3.8-3.

Thank
You!